

GG2037

Introduction to Geoinformatics

Subject Area: GG - Geography**Credit Weighting:** 5*The notional student workload for 5 credits is 125 hours***Semester(s)**

Semester 2 (Jan-May)

No. of Students

Min 10, Max 120

Pre-requisite(s)

None

Co-requisite(s)

None

Teaching Methods

- 24 x 1 Hr(s) Lectures

Module Co-ordinator

Dr Maedhbh Nic Lochlainn - Department of Geography

Lecturer(s)

- Dr Fiona Cawkwell - Department of Geography
- Dr Maedhbh Nic Lochlainn - Department of Geography
- Staff - Department of Geography

Guest Lecturer(s)

None

Module Goal

To give students an introduction to the key concepts, principles and theory of GIS, Remote Sensing, global satellite navigation systems, and related geoinformatics technologies, their use in Geographical, Earth and Environmental sciences, and practical techniques of spatial data analysis.

Module Content

Integrated lectures and practical exercises are designed to introduce students to techniques and systems for geospatial information display, manipulation and analysis. Concepts of spatial data, databases, data models, data sources, data acquisition and image analysis are explored in lectures and reinforced by computer-based assignments. Applications of GIS and Remote Sensing within Geographical, Earth and Environmental sciences and for supporting the work of the UN Sustainable Development Goals are considered.

Learning Outcomes

On successful completion of this module, students should be able to:

1. Explain fundamental GIS principles, including the components of a GIS, coordinate systems, spatial data models, spatial databases, and topological relationships, as covered in lectures and readings for the module.
2. Perform basic spatial analysis on a range of geographic and environmental topics in computer-based practicals.
3. Describe the electromagnetic spectrum and how radiation of different wavelengths interacts with features on the Earth's surface, as covered in lectures and readings for the module.
4. Interpret digital images acquired by different sensors and apply image processing techniques in order to enhance the quality of information derived from images in computer-based practicals.
5. Critically evaluate how Geoinformatics can be used to support the work of the Un Sustainable Development Goals using content from lectures, readings, and computer-based practicals.
6. Use standard, commercial Windows-based and online Geoinformatics software in computer-based practicals.

Examination and Assessment

Total Marks 100

- Continuous Assessment 100 Marks
 - Assignment - 4 practical exercises 20 marks each, 1 x MCQ 20 marks (100 Marks)

Penalties (for late submission of Course/Project Work etc.)

Where work is submitted up to and including 7 days late, 5% of the total marks available shall be deducted from the mark achieved. Where work is submitted up to and including 14 days late, 10% of the total marks available shall be deducted from the mark achieved. Work submitted 15 days late or more shall be assigned a mark of zero

Pass Standard

40%

Special Requirements for Passing Module

None

Supplemental Examination and Assessment

- Continuous Assessment
 - Marks in passed elements of continuous assessment are carried forward. Failed elements must be repeated
 - Assessments as prescribed by the Department whereby an element constitutes the GIS practicals, the RS practicals, and the MCQ.